Evaluating the Robustness of Computer Vision Models

Project by: Ayelet & Noa

BDD100K road scenes · YOLOv8 · Mask R-CNN · MiDaS

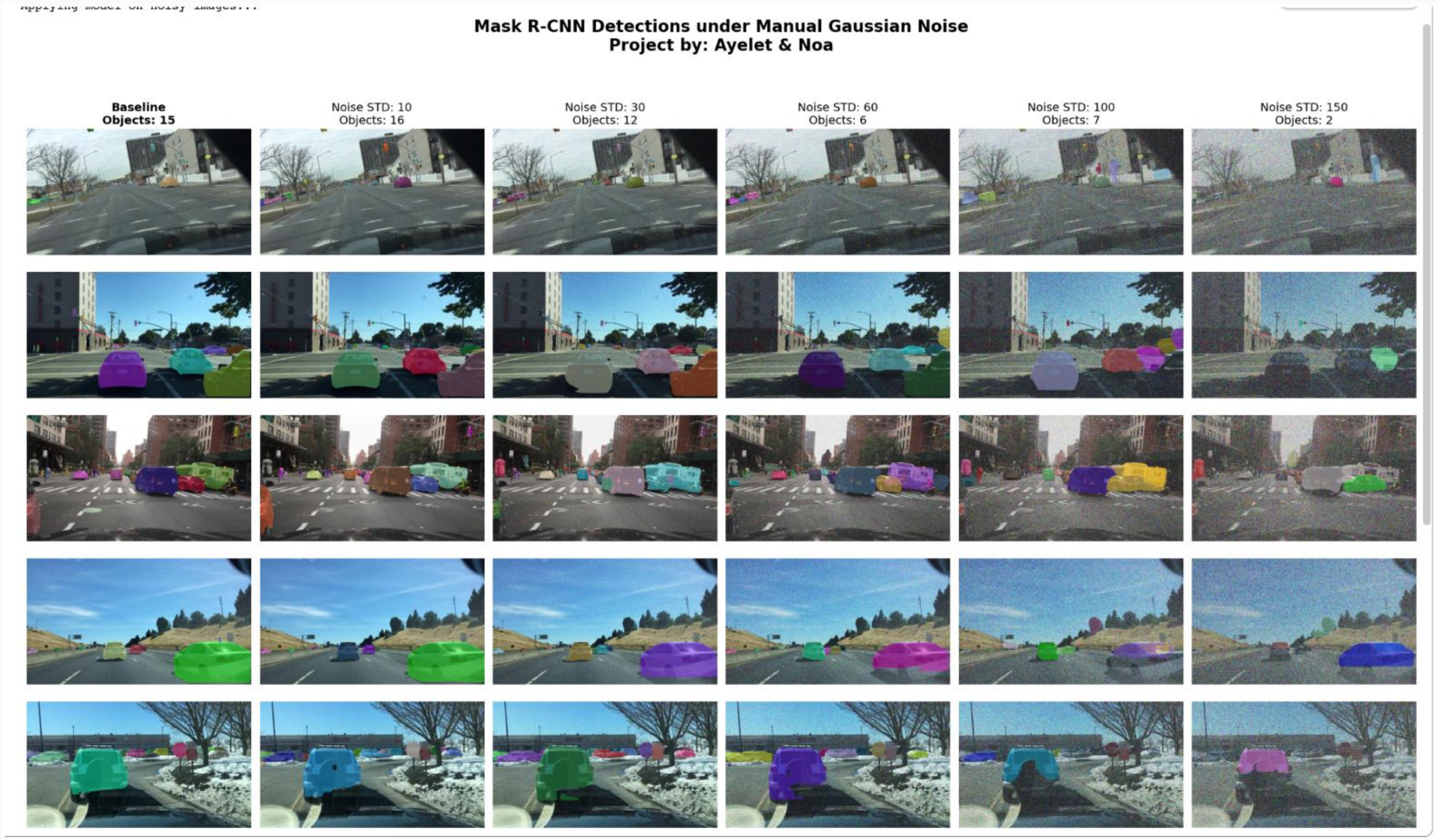
Project Goal

Measure how vision models behave when image quality drops
Test three tasks: object detection, instance segmentation and depth estimation
Compare clean baselines, classical enhancement and fine-tuning
Identify when simple filtering is enough and when model adaptation is needed

Example: gradual low-light degradation

Evaluation Method

Start from clean images and define a baseline output
Apply one distortion at several severity levels
Run the model again and compare against the clean baseline
Use Recall and Precision (when needed)
for detection and segmentation Use RMSE and SSIM for depth-map comparison

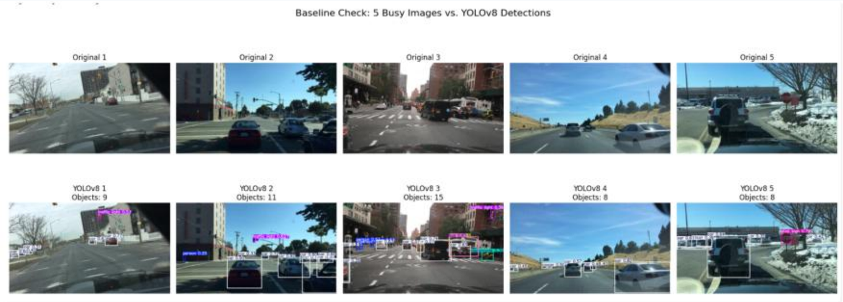


Clean baseline outputs were used as the reference for later comparisons

Models and Metrics

YOLOv8

Object detection
Recall vs. distortion level
Confidence loss and misclassification

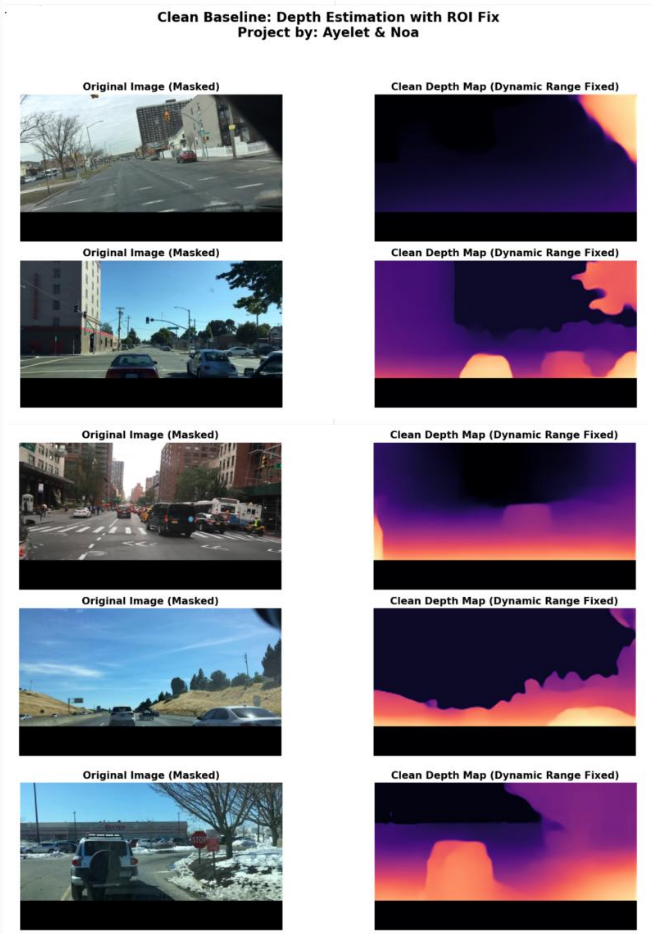


Mask R-CNN

Instance segmentation
Recall and Precision
Effect of masks under noise and compression

MiDaS

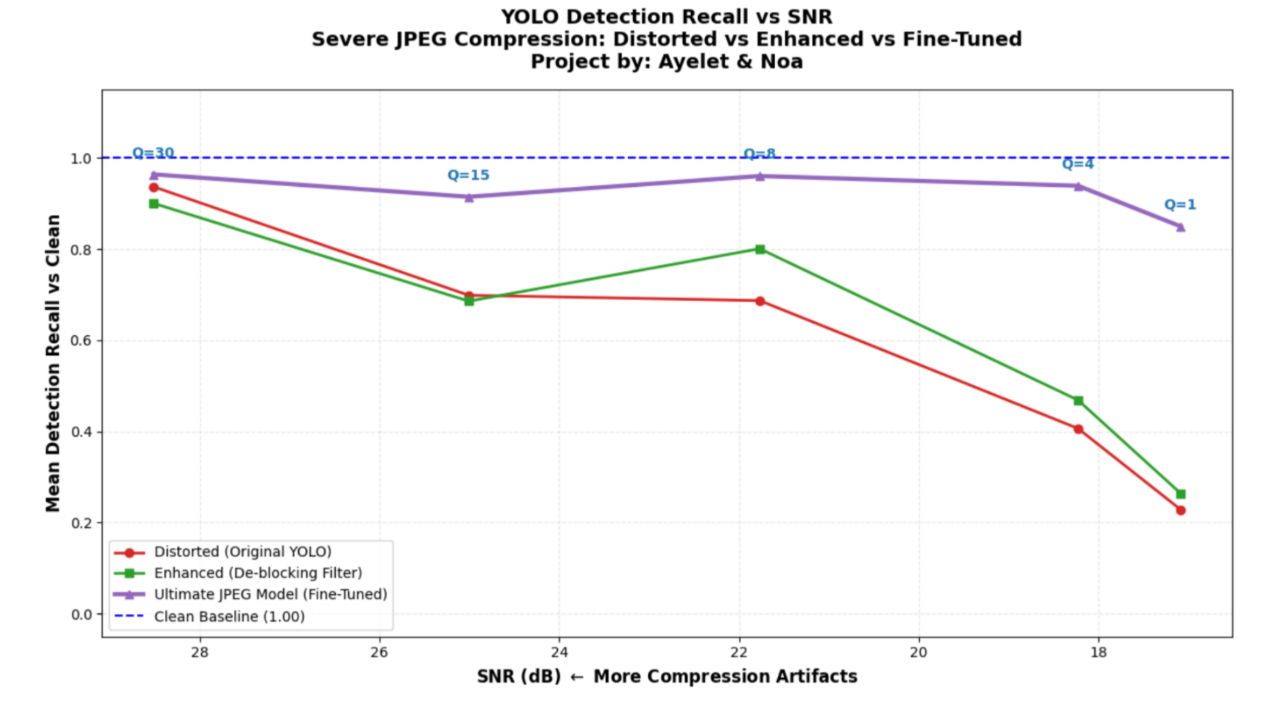
Monocular depth estimation
RMSE, lower is better
SSIM, higher is better

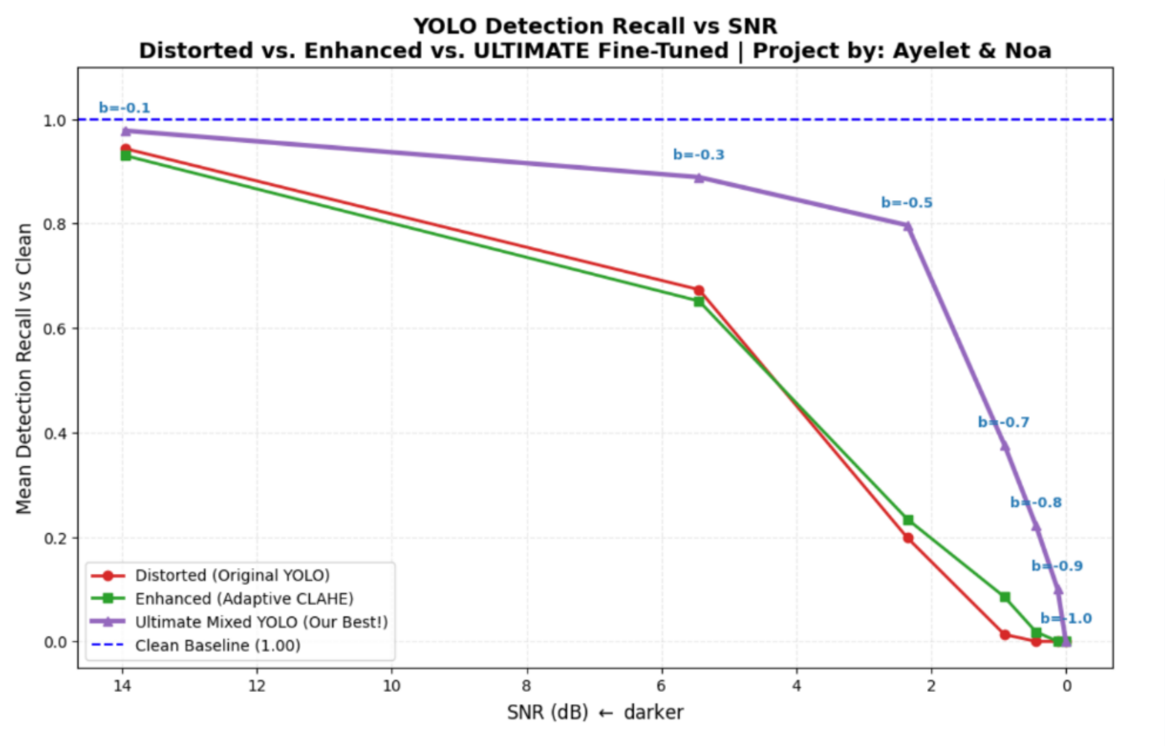


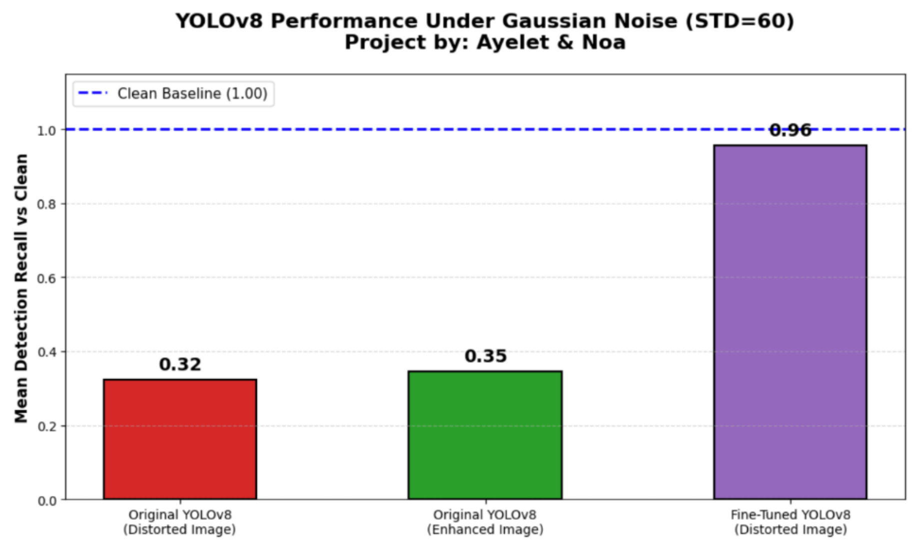
One dataset, three tasks, separate metrics

YOLOv8: Detection Robustness

Low light gradually removes visible objects from the detector
Gaussian noise reduces feature stability and confidence
JPEG artifacts may create false object-like structures
Fine-tuning improves recall under difficult distortions, but the original model can remain best under mild distortion



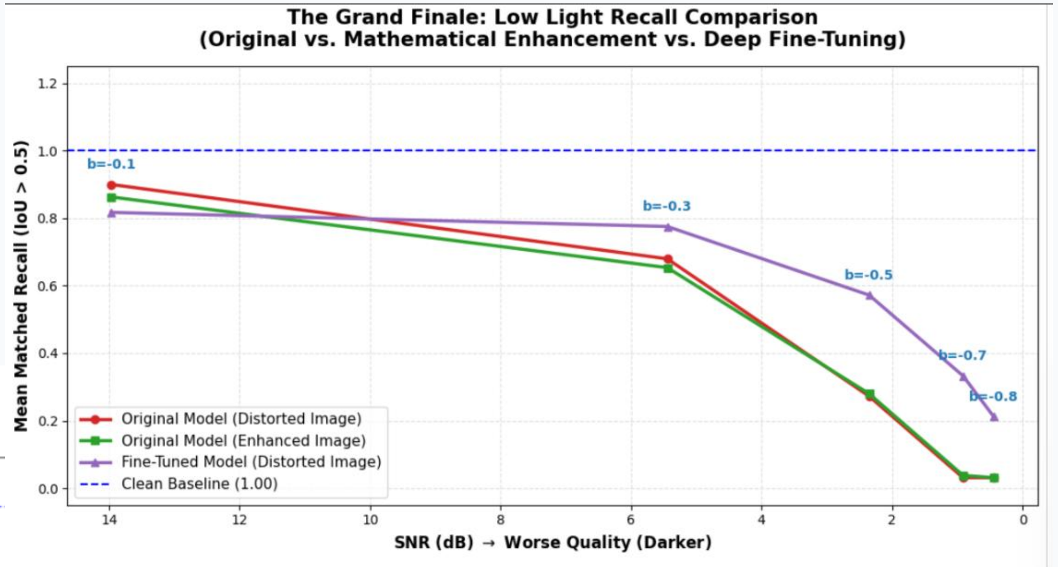


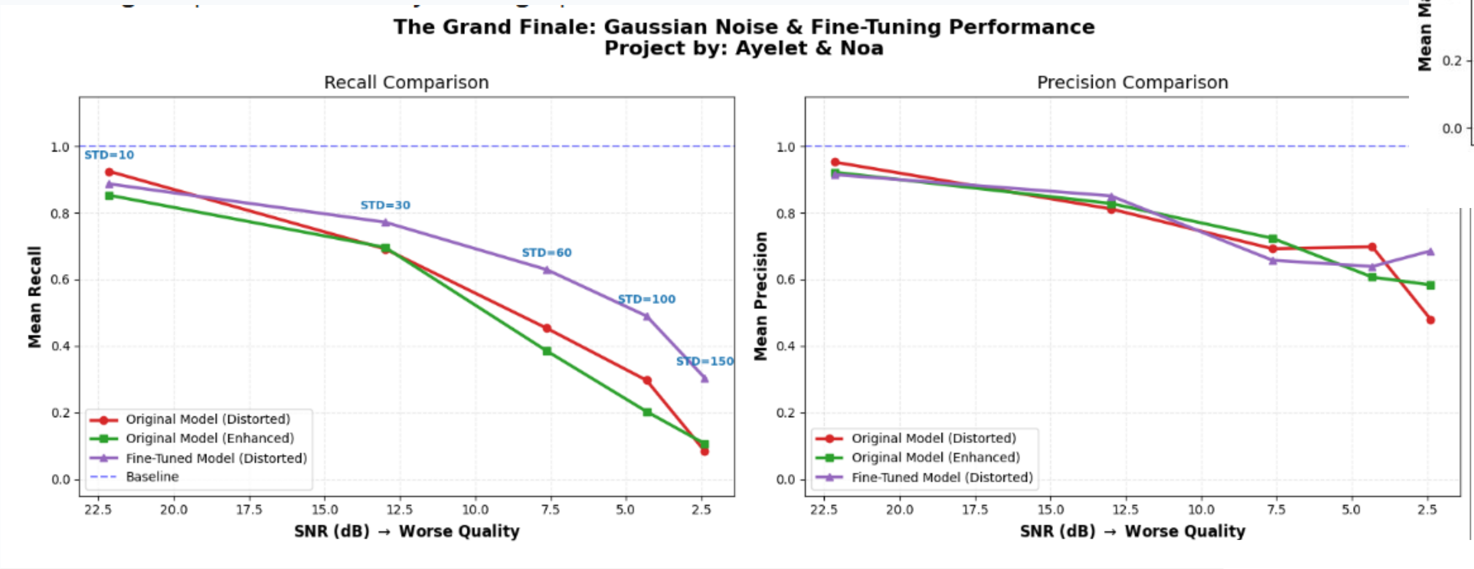


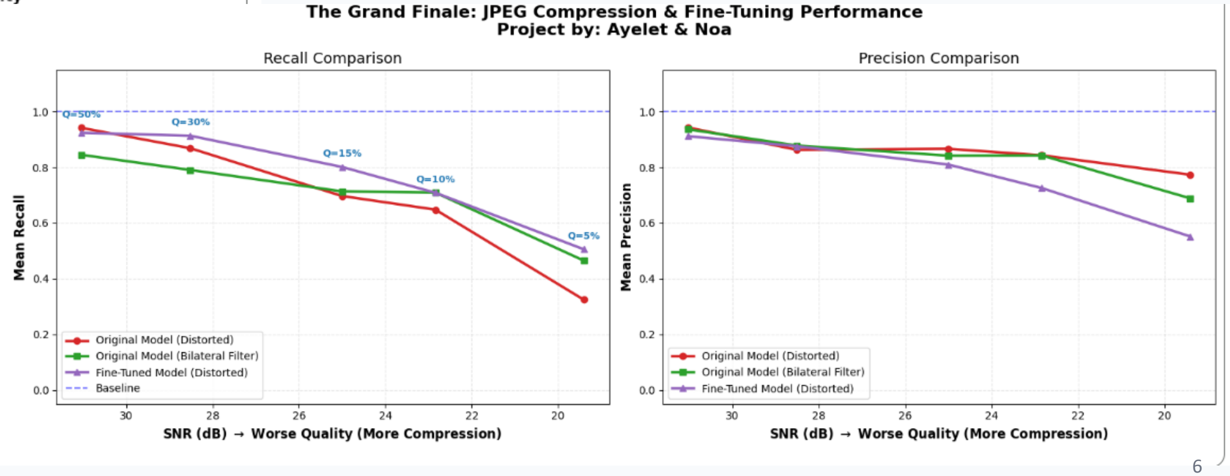
YOLOv8 results show a clear trade-off between clean accuracy and robustness

Mask R-CNN: Segmentation Robustness

Instance masks become unstable when textures are corrupted
Classical filters can help, but the effect depends on the distortion
Fine-tuning was trained with pseudo-labels generated from clean images
At extreme noise and compression, the fine-tuned model preserved higher recall



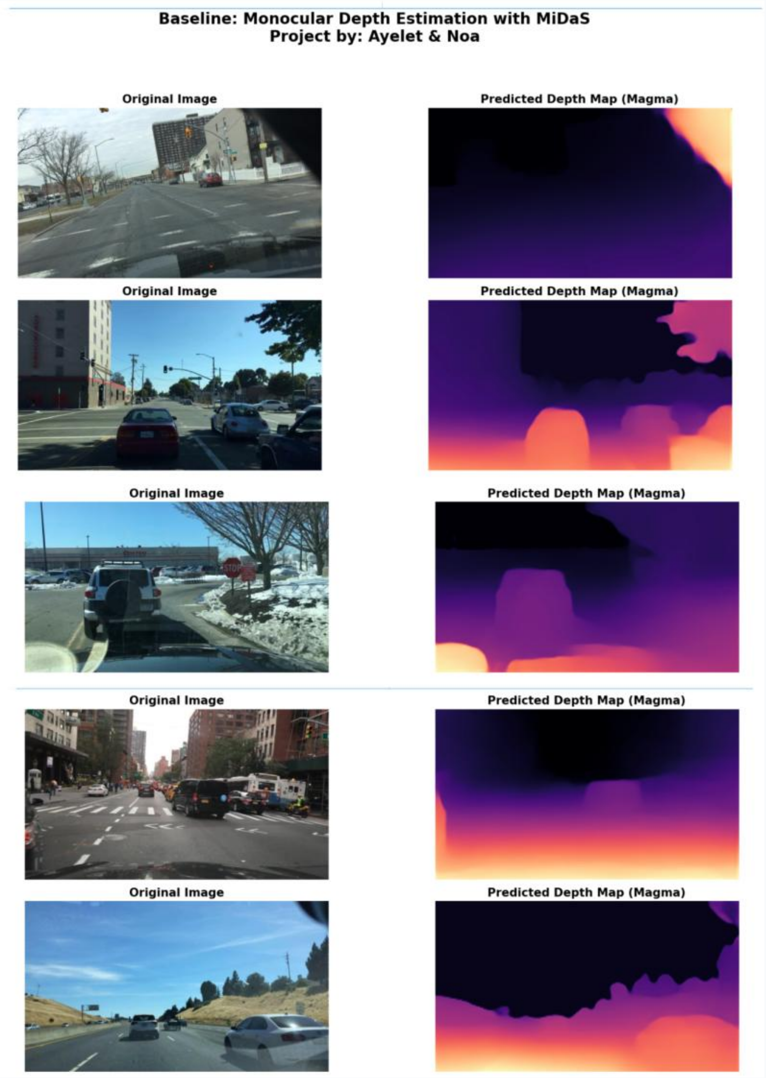


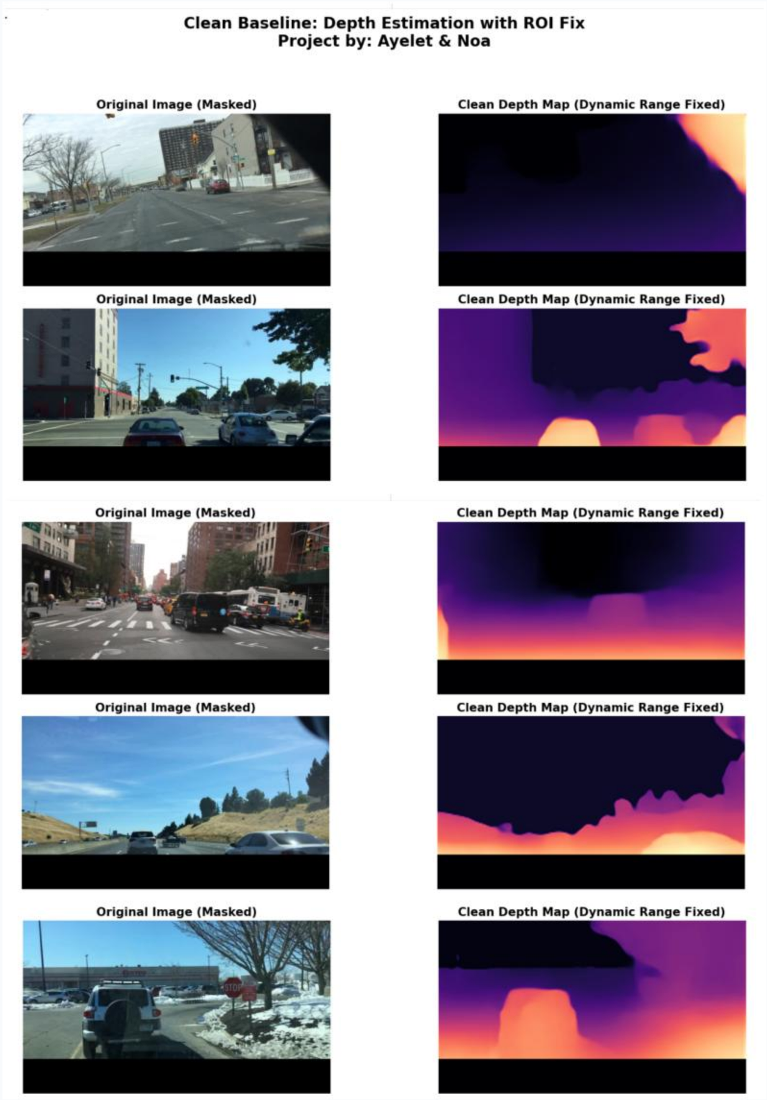


Fine-tuning compared against the original model and classical enhancement

MiDaS: Depth Estimation Setup

MiDaS predicts a relative depth map from a single image
The dashboard region was removed from evaluation using an upper 80% ROI
Depth maps were normalized before comparison
RMSE captures numeric error; SSIM captures structural similarity



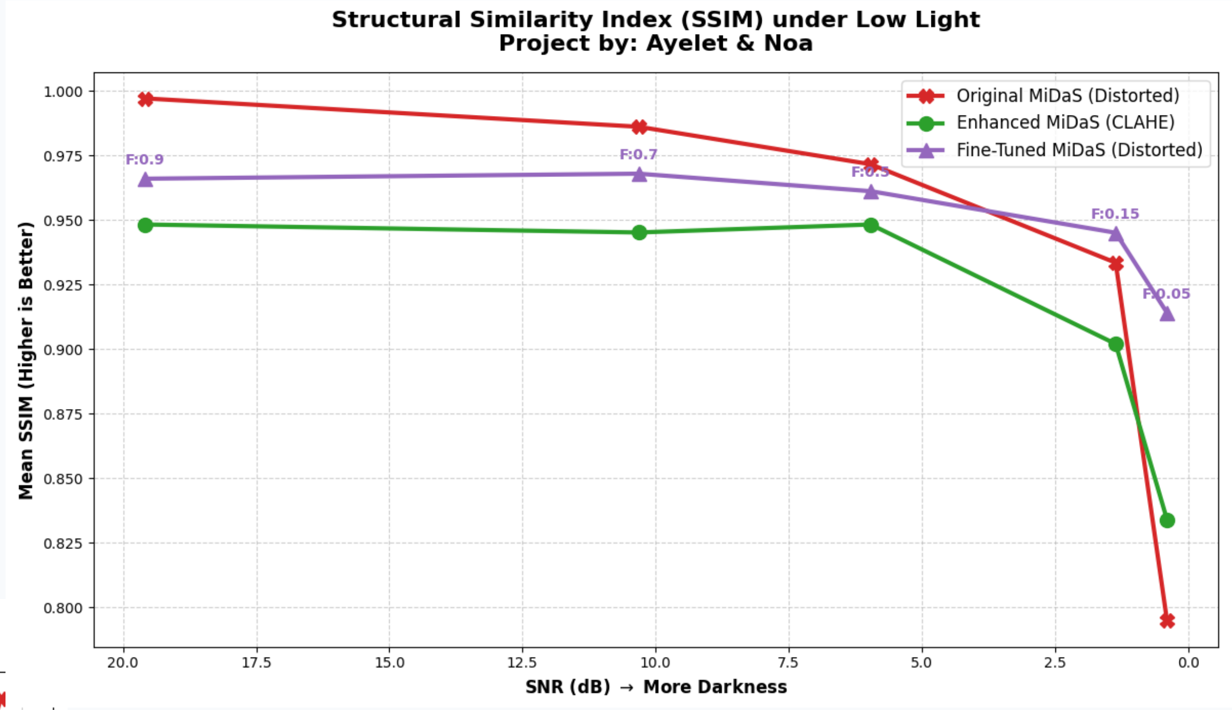


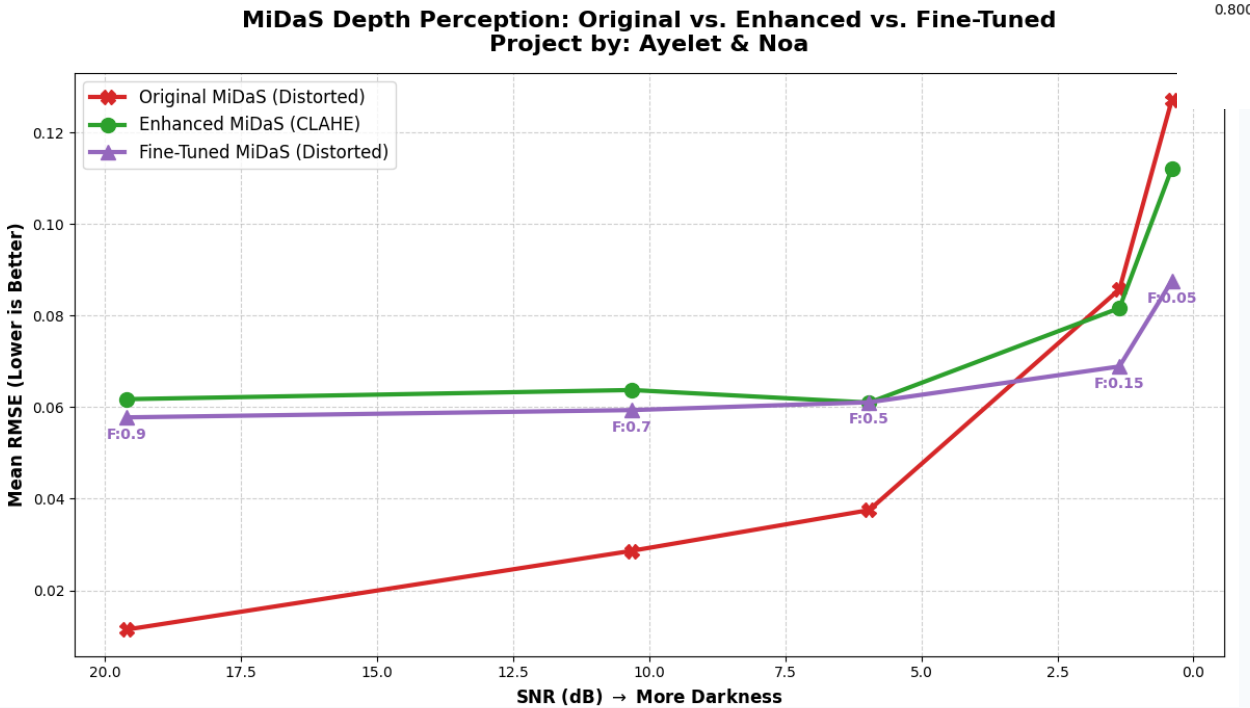
Baseline depth maps before and after ROI correction

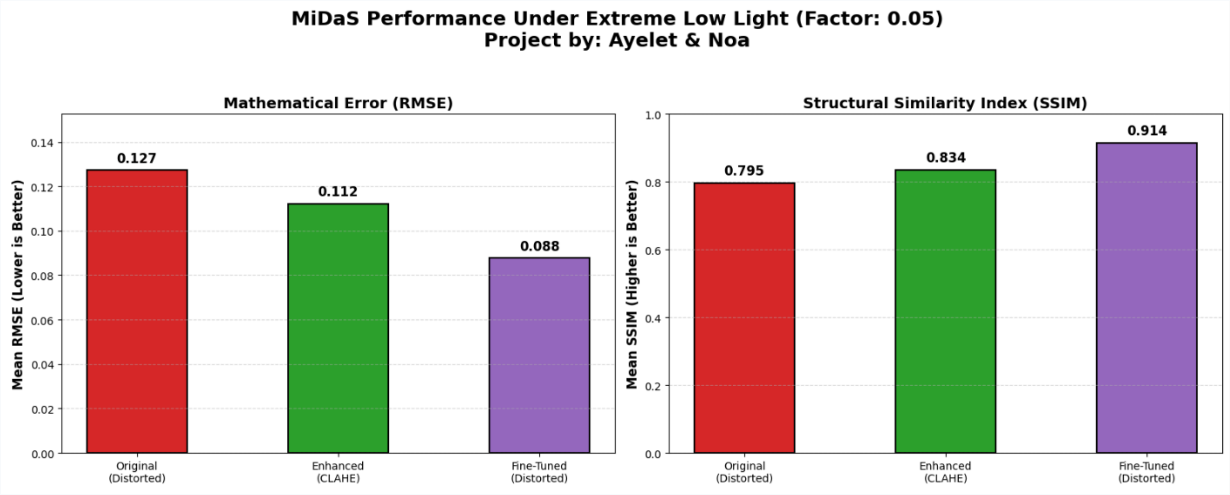
MiDaS under Low Light

Darkness weakens the visual signal used for depth estimation

RMSE increases as the image becomes darker
SSIM decreases because the depth structure changes
CLAHE helps mainly under strong darkness
Fine-tuning gives the most reliable recovery at extreme low light



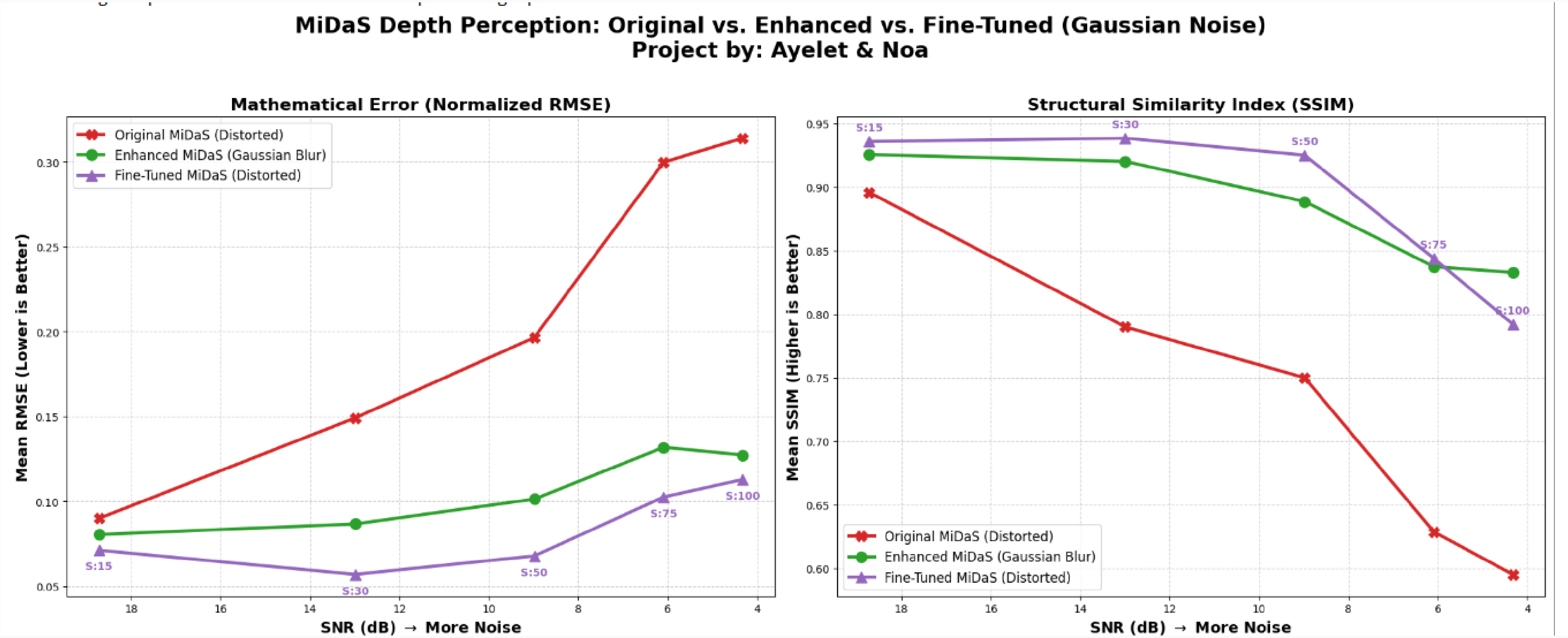




Fine-tuned MiDaS improves both RMSE and SSIM at the darkest level

MiDaS under Gaussian Noise

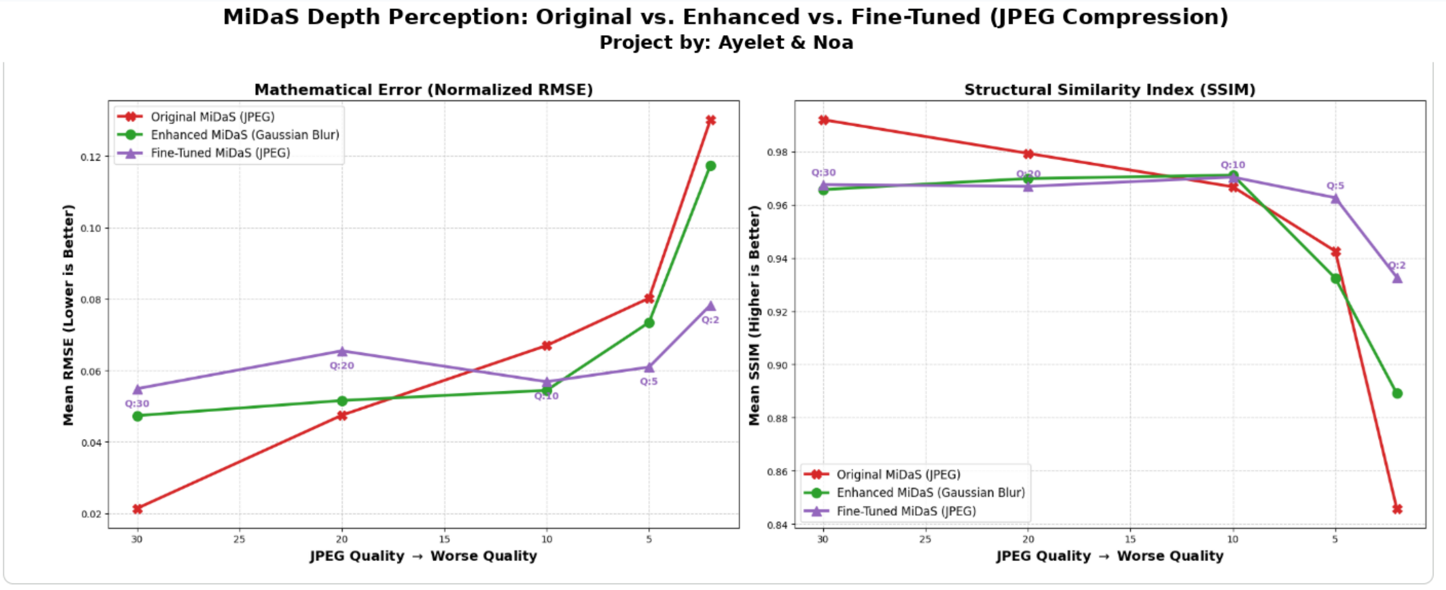
The noisy baseline shows rising RMSE and falling SSIM
Gaussian Blur improves results at every noise level
Fine-tuning achieves the lowest RMSE across all tested noise levels
At very high noise, blur can retain slightly smoother structure

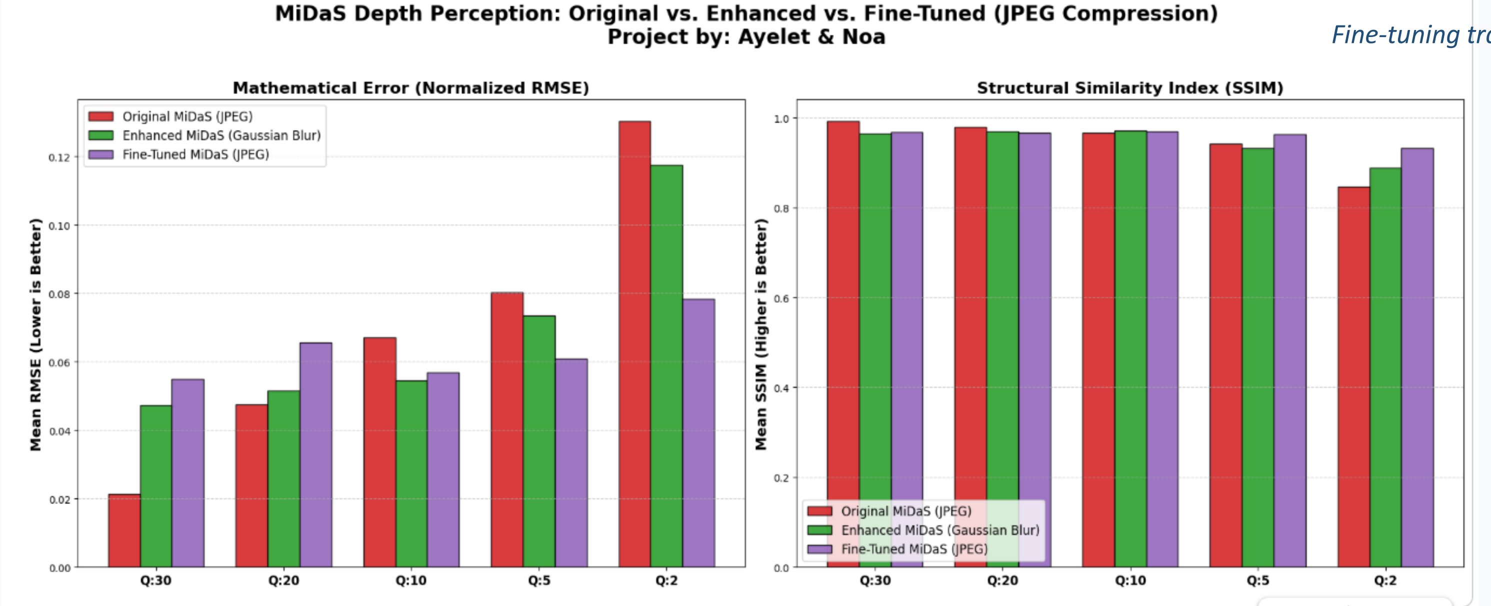


Gaussian Blur helps consistently; fine-tuning focuses on lowering numeric depth error

MiDaS under JPEG Compression

The original model is strongest when compression is mild
Gaussian Blur partially reduces blocking artifacts
Fine-tuning becomes clearly better at low quality factors
At Q=2, the fine-tuned model gives the lowest RMSE and highest SSIM

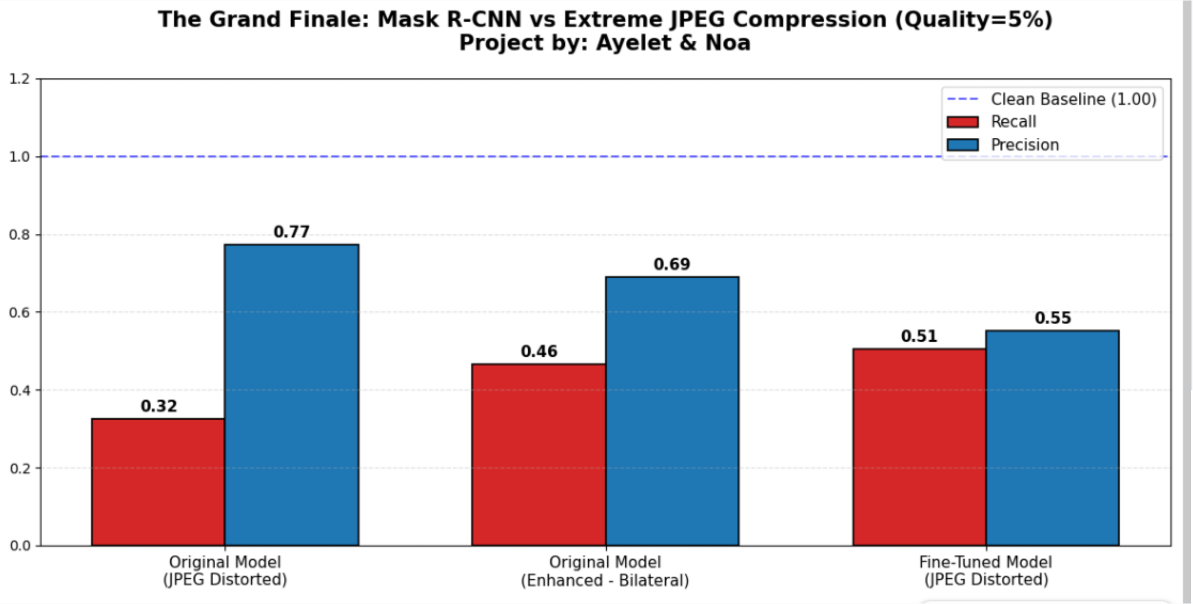


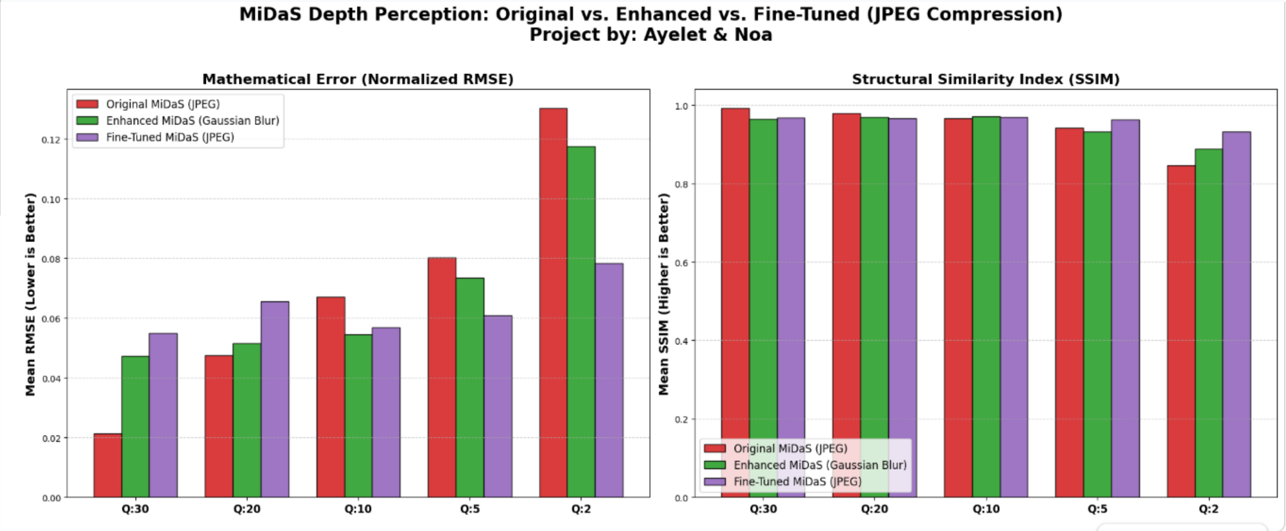


Fine-tuning trades a small loss at high quality for stronger robustness at severe compression

What We Learned

Model robustness depends strongly on the distortion type
Classical enhancement is useful when the distortion has a simple signal-processing structure
Fine-tuning is more reliable when distortions become severe or structural
A single metric is not enough: recall, precision, RMSE and SSIM can tell different stories
The best system should choose between original, enhanced and fine-tuned models according to image quality





Final results highlight the advantage of fine-tuning under severe degradation

Conclusion

Computer vision models are reliable on clean images, but performance drops under realistic distortions
Classical enhancement is simple and useful, especially for Gaussian noise
Fine-tuning provides stronger robustness under severe low light, noise and JPEG compression
Robust real-world systems should combine quality assessment, preprocessing and model adaptation

This course was interesting and we really enjoyed,

Thank you ☺

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